

Consolidated Power Quality Event Classification Results

1. Optimal Feature Selection Summary

The following feature pairs were mathematically selected as the single best common pair for 4-class multi-class rule-based classification across each respective dataset:

- **CST Dataset:** F3 and F4
- **BGST Dataset:** F1 and F5
- **HST Dataset:** F1 and F5

2. Rules Used for Each Type (Base Classifier)

CST Dataset (F3 & F4)

- **Simulated:** IF F3 <= 0.00012364
- **Surface:** ELSE IF F3 <= 0.000562287 AND F4 <= 10.1456
- **Void:** ELSE IF F3 <= 0.000562287 AND F4 > 10.1456
- **Sharp Point:** ELSE IF F3 > 0.000562287 AND F4 <= 12.6887
- **Void (Fallback):** ELSE IF F3 > 0.000562287 AND F4 > 12.6887

BGST Dataset (F1 & F5)

- **Simulated:** IF F1 <= 7.89e-05
- **Surface:** ELSE IF F1 <= 0.0003135 AND F5 <= 197.5
- **Void:** ELSE IF F1 <= 0.0003135 AND F5 > 197.5
- **Surface (Fallback):** ELSE IF F1 > 0.0003135 AND F5 <= 65.4116
- **Sharp Point:** ELSE IF F1 > 0.0003135 AND F5 > 65.4116

HST Dataset (F1 & F5)

- **Simulated:** IF F1 <= 7.41e-05
- **Surface:** ELSE IF F1 <= 0.0003325 AND F5 <= 200.5
- **Void:** ELSE IF F1 <= 0.0003325 AND F5 > 200.5
- **Sharp Point:** ELSE IF F1 > 0.0003325

3. CST Dataset Classification Tables

POWER QUALITY EVENT CLASSIFICATION RESULTS (Reference Condition)

Power Quality Event	Correct	Wrong	% Success
Simulated	400	0	100.00
Sharp Point	112	4	96.55
Void	205	48	81.03

POWER QUALITY EVENT CLASSIFICATION RESULTS (20 dB Condition)

Power Quality Event	Correct	Wrong	% Success
Simulated	400	0	100.00
Sharp Point	0	253	0.00
Void	93	160	36.76
Surface	250	3	98.81
Total	743	416	64.11